Scientific Notation

Name _____ **Date** _____ No calculator **Version 1**

KEY WORDS scientific notation · standard form · power of ten · exponent · decimal point

A Large numbers ●○○ — ANSWER ON THE SHEET

Remember: Scientific notation is a number from 1 up to (but not including) 10, times a power of ten: $62{,}000 = 6.2 \times 10^4$.

1. Write each number in scientific notation.

a) 5,000 = _____ **b)** 700,000 = _____ **c)** 62,000 = _____

d) 380 = _____ **e)** 9,100,000 = _____ **f)** 4,750 = _____

g) 106,000 = _____ **h)** 83,000,000 = _____ **i)** 2,500,000,000 = _____

j) 19 = _____ **k)** 60,400 = _____ **l)** 3,330,000 = _____

B Small numbers ●●○ — ANSWER ON THE SHEET

Remember: A number smaller than 1 has a negative power of ten: $0.0045 = 4.5 \times 10^{-3}$.

2. Write each number in scientific notation.

a) 0.03 = _____ **b)** 0.0008 = _____ **c)** 0.0045 = _____

d) 0.71 = _____ **e)** 0.000026 = _____ **f)** 0.00904 = _____

g) 0.0000012 = _____ **h)** 0.055 = _____ **i)** 0.000638 = _____

j) 0.9 = _____ **k)** 0.0101 = _____ **l)** 0.00000077 = _____

C Back to ordinary numbers ●●○ — ANSWER ON THE SHEET

Remember: A positive power makes the number larger. A negative power makes it smaller.

3. Write each number as an ordinary number.

a) 4×10^2 = _____ **b)** 6.3×10^4 = _____ **c)** 2.8×10^6 = _____

d) 9×10^{-2} = _____ **e)** 5.1×10^{-3} = _____ **f)** 7.25×10^3 = _____

g) 1.4×10^{-5} = _____ **h)** 3.06×10^5 = _____ **i)** 8×10^{-1} = _____

j) 9.9×10^1 = _____ **k)** 2.5×10^{-4} = _____ **l)** 1.234×10^7 = _____

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D Comparing ●●○ — WORK IN YOUR BOOK

Remember: Compare the powers of ten first. If they are the same, compare the numbers in front.

4. Answer each question.

a) Put these in order, smallest first: 4.2×10^5, 9.8×10^3, 1.5×10^6, 7×10^4.

b) Put these in order, smallest first: 3.1×10^{-2}, 6.5×10^{-4}, 2×10^{-1}, 8.8×10^{-3}.

c) Put these in order, smallest first: 5×10^3, 5×10^{-3}, 2.4×10^0, 9.9×10^2.

d) Put these in order, smallest first: 6.1×10^7, 6.01×10^7, 9.9×10^6, 1.1×10^8.

e) Put these in order, smallest first: 4×10^{-5}, 3.9×10^{-5}, 4.1×10^{-6}, 1×10^{-4}.

E Calculating ●●● — WORK IN YOUR BOOK

Remember: Multiply or divide the numbers in front, and add or subtract the powers of ten. Then check that the answer is in proper scientific notation.

5. Work out. Give each answer in scientific notation.

a) $(2 \times 10^3) \times (4 \times 10^5)$

b) $(3 \times 10^4) \times (3 \times 10^2)$

c) $(8 \times 10^6) \div (2 \times 10^2)$

d) $(9 \times 10^5) \div (3 \times 10^3)$

e) $(4 \times 10^3) \times (5 \times 10^4)$

f) $(6 \times 10^{-2}) \times (3 \times 10^5)$

g) $(2 \times 10^7) \div (8 \times 10^3)$

h) $(1.5 \times 10^4) \times (4 \times 10^{-6})$

i) $(9 \times 10^8) \div (3 \times 10^{-2})$

j) $(2.5 \times 10^3) \times (2 \times 10^3)$

★ Extension — FOR EARLY FINISHERS

6. These are not in proper scientific notation. Write each one correctly.

a) 34×10^3 **b)** 0.6×10^5 **c)** 250×10^{-4}

7. The distance from the Earth to the Sun is about 150,000,000 km. Write this distance in scientific notation.

8. A computer does 4×10^6 calculations each second. How many calculations does it do in 50 seconds? Give your answer in scientific notation.

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Cell Tuition
1-1 PRIVATE TUTORS

YEAR 9 · NUMBER

Scientific Notation

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS scientific notation · standard form · power of ten · exponent · decimal point

A Large numbers ●○○

ANSWER ON THE SHEET

Remember: Scientific notation is a number from 1 up to (but not including) 10, times a power of ten: $62{,}000 = 6.2 \times 10^4$.

1. Write each number in scientific notation.

a) 4,000 = _____ **b)** 900,000 = _____ **c)** 36,000 = _____

d) 720 = _____ **e)** 5,400,000 = _____ **f)** 8,250 = _____

g) 209,000 = _____ **h)** 67,000,000 = _____ **i)** 1,500,000,000 = _____

j) 28 = _____ **k)** 70,300 = _____ **l)** 4,440,000 = _____

B Small numbers ●●○

ANSWER ON THE SHEET

Remember: A number smaller than 1 has a negative power of ten: $0.0045 = 4.5 \times 10^{-3}$.

2. Write each number in scientific notation.

a) 0.05 = _____ **b)** 0.0006 = _____ **c)** 0.0025 = _____

d) 0.39 = _____ **e)** 0.000084 = _____ **f)** 0.00702 = _____

g) 0.0000046 = _____ **h)** 0.015 = _____ **i)** 0.000374 = _____

j) 0.4 = _____ **k)** 0.0202 = _____ **l)** 0.00000088 = _____

C Back to ordinary numbers ●●○

ANSWER ON THE SHEET

Remember: A positive power makes the number larger. A negative power makes it smaller.

3. Write each number as an ordinary number.

a) 7×10^2 = _____ **b)** 4.9×10^4 = _____ **c)** 3.2×10^6 = _____

d) 6×10^{-2} = _____ **e)** 8.7×10^{-3} = _____ **f)** 2.15×10^3 = _____

g) 9.6×10^{-5} = _____ **h)** 5.08×10^5 = _____ **i)** 3×10^{-1} = _____

j) 6.6×10^1 = _____ **k)** 7.5×10^{-4} = _____ **l)** 5.678×10^7 = _____

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D Comparing ●●○ — WORK IN YOUR BOOK

Remember: Compare the powers of ten first. If they are the same, compare the numbers in front.

4. Answer each question.

a) Put these in order, smallest first: 3.6×10^5, 8.1×10^3, 2.5×10^6, 6×10^4.

b) Put these in order, smallest first: 4.7×10^{-2}, 9.2×10^{-4}, 3×10^{-1}, 5.5×10^{-3}.

c) Put these in order, smallest first: 7×10^3, 7×10^{-3}, 1.8×10^0, 9.5×10^2.

d) Put these in order, smallest first: 5.2×10^7, 5.02×10^7, 9.8×10^6, 1.2×10^8.

e) Put these in order, smallest first: 6×10^{-5}, 5.9×10^{-5}, 6.1×10^{-6}, 1×10^{-4}.

E Calculating ●●● — WORK IN YOUR BOOK

Remember: Multiply or divide the numbers in front, and add or subtract the powers of ten. Then check that the answer is in proper scientific notation.

5. Work out. Give each answer in scientific notation.

a) $(3 \times 10^3) \times (2 \times 10^5)$

b) $(2 \times 10^4) \times (4 \times 10^2)$

c) $(6 \times 10^6) \div (3 \times 10^2)$

d) $(8 \times 10^5) \div (4 \times 10^3)$

e) $(5 \times 10^3) \times (6 \times 10^4)$

f) $(7 \times 10^{-2}) \times (2 \times 10^5)$

g) $(3 \times 10^7) \div (6 \times 10^3)$

h) $(2.5 \times 10^4) \times (2 \times 10^{-6})$

i) $(8 \times 10^8) \div (2 \times 10^{-2})$

j) $(1.5 \times 10^3) \times (6 \times 10^3)$

★ Extension — FOR EARLY FINISHERS

6. These are not in proper scientific notation. Write each one correctly.

a) 48×10^3 **b)** 0.9×10^5 **c)** 360×10^{-4}

7. Light travels about 300,000 km in one second. Write this distance in scientific notation.

8. A factory makes 3×10^4 bottles each day. How many bottles does it make in 200 days? Give your answer in scientific notation.

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